99 — Canonical Reading Index, Primary Citations, and Bibliographic Registry

Project Title: Research, Design, and Implementation of an Optimized Gomoku Game Engine with Artificial Intelligence **Academic Module:** Basic Project (■■ Án C■ S■) — Software Engineering & AI Track **Document Standard:** Standard 00 (IMRaD & Evidence Specification for Software Systems) **Control Layer ID:** `REF-INDEX-99` **Language:** English **Status:** Comprehensive Academic Bibliography

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1. Bibliographic Taxonomy & Metadata Schema

Each bibliographic entry is classified under strict academic metadata: ***Reference ID:*** *Unique citation identifier*. **Type:** Primary Academic Paper, Doctoral Thesis, Technical Specification, Official Book. ***Read Status:*** `FULL_READ`, `RELEVANT_SECTION_READ`, `CITED`. **Used By:** Specific research questions (`RQ`) and architecture decision records (`ADR`).

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2. Canonical Master References

REF-01: Victor Allis (1994) — Gomoku Solvability Proof

- **Citation:** Allis, L. V. (1994). *Searching for Solutions in Games and Artificial Intelligence*. Ph.D. Thesis, University of Limburg, Maastricht.
- **Type:** Academic Doctoral Dissertation (Primary Source).
- **Official URL:** https://www.cs.unimaas.nl/olympiad/thesis/Allis_thesis.pdf
- **Read Status:** `FULL_READ`.
- **Used By:** `RQ-W1-01`, `RQ-W4-B01`, `RQ-W7-01`, Thesis Chapters 1, 2.
- **Key Findings:** Free-style 15×15 Gomoku is mathematically proven to be a forced win for the first player (Black) using Proof-Number Search and Threat-Space Search.

REF-02: Donald E. Knuth & Ronald W. Moore (1975) — Alpha-Beta Pruning

- **Citation:** Knuth, D. E., & Moore, R. W. (1975). *An Analysis of Alpha-Beta Pruning*. Artificial Intelligence, 6(4), 293–326.
- **Type:** Peer-Reviewed Journal Article (Primary Source).
- **Official DOI:** https://doi.org/10.1016/0004-3702(75)90019-3
- **Read Status:** `FULL_READ`.
- **Used By:** `RQ-W6-01`, `RQ-W8-01`, Thesis Chapters 2, 4, 6.
- **Key Findings:** Formal mathematical proof of the Alpha-Beta pruning algorithm. Established that optimal move ordering reduces tree complexity to $\mathcal{O}(b^{d/2})$.

REF-03: Albert L. Zobrist (1970) — Zobrist Hashing

- **Citation:** Zobrist, A. L. (1970). *A New Hashing Method with Application for Game Playing*. Technical Report #88, Computer Sciences Department, University of Wisconsin, Madison.
- **Type:** University Technical Report (Primary Source).
- **Official Archive:** https://minds.wisconsin.edu/handle/1793/57606
- **Read Status:** `FULL_READ`.
- **Used By:** `RQ-W9-01`, `RQ-W9-02`, Thesis Chapters 2, 4, 5.

- **Key Findings:** Introduced bitwise XOR hashing for discrete board games, enabling $O(1)$ state updates during move/undo cycles.

REF-04: Claude E. Shannon (1950) — Game Tree Search Paradigm

- **Citation:** Shannon, C. E. (1950). *Programming a Computer for Playing Chess.* Philosophical Magazine, Ser.7, Vol. 41, No. 314, 256–275.
- **Type:** Foundational Journal Article.
- **Official Archive:** https://vision.unipv.it/IA1/programmingacomputer.pdf
- **Read Status:** `FULL_READ`.
- **Used By:** `RQ-W5-01`, `RQ-W4-C01`, Thesis Chapters 1, 2.
- **Key Findings:** Established the minimax search framework, heuristic evaluation concept, and type-A vs. type-B lookahead strategies.

REF-05: John F. Nash (1950) — Strategy-Stealing Argument

- **Citation:** Nash, J. F. (1950). *Non-Cooperative Games.* Annals of Mathematics, 54(2), 286–295.
- **Official DOI:** https://doi.org/10.2307/1969529
- **Type:** Mathematical Paper.
- **Read Status:** `RELEVANT_SECTION_READ`.
- **Used By:** `RQ-W1-01`, Thesis Chapter 2.
- **Key Findings:** Proved that in symmetric, zero-sum combinatorial games with no zugzwang, the second player can never have a forced winning strategy.

REF-06: Gomocup International Tournament Consortium (2000–2025)

- **Citation:** Gomocup International Tournament Consortium. (2024). *Gomocup AI Protocol Specifications and Tournament Rules.*
- **Type:** Official Tournament Standard.
- **Official URL:** https://gomocup.org/detail-information/
- **Read Status:** `FULL_READ`.
- **Used By:** `RQ-ARCH-01`, `RQ-W10-04`, Thesis Chapters 1, 3.
- **Key Findings:** Standardized inter-process communication protocol (`START`, `TURN`, `BOARD`, `INFO`, `END`) for Gomoku AI programs.

REF-07: W3C WebAssembly Working Group (2024)

- **Citation:** Rossberg, A., et al. (2024). *WebAssembly Core Specification 2.0.* W3C Recommendation.
- **Type:** Official International Standard.
- **Official URL:** https://www.w3.org/TR/wasm-core-2/
- **Read Status:** `FULL_READ`.
- **Used By:** `ADR-ARCH-01`, `RQ-W4-A01`, Thesis Chapters 3, 5.
- **Key Findings:** Defines the binary execution environment, structured control flow, and linear memory operations for client-side web computing.

REF-08: WHATWG Web Workers Standard (2024)

- **Citation:** WHATWG. (2024). *HTML Living Standard: Web Workers.*
- **Type:** Official Web Standard.
- **Official URL:** https://html.spec.whatwg.org/multipage/workers.html
- **Read Status:** `FULL_READ`.

- **Used By:** `ADR-ARCH-01`, `RQ-W3-01`, Thesis Chapters 3, 4.
- **Key Findings:** Defines background execution contexts and message-passing protocols for multithreaded web applications.